

Business Problem Statement

“Loan approval decisions are critical for both financial institutions and applicants, yet many approvals are influenced by complex factors spanning income, credit history, employment, and demographics. Financial institutions need data-driven insights to identify which applicant characteristics most strongly impact loan approval decisions. This project analyzes a dataset of loan applications from the US and Canada to uncover patterns, build predictive understanding, and provide actionable recommendations to improve approval accuracy and risk assessment.”

Deliverables (Project Workflow)

1. Data Import & Initial Exploration

- Import the loan approval dataset into Python.
- Explore the dataset structure, data types, and completeness.
- Conduct an initial review to understand variables like applicant income, credit score, employment, loan amount, and approval status.

2. Data Cleaning & Preparation

- Handle missing values, inconsistencies, and incorrect data types.
- Standardize categorical variables and encode them if necessary.
- Create derived features (e.g., debt-to-income ratio, credit tiers) that may help analysis.
- Remove irrelevant or redundant columns to make the dataset ready for analysis.

3. Data Transformation & Organization

- Structure the cleaned dataset for analytical tasks, including SQL queries or predictive modeling.
- Convert categorical variables into numerical formats where needed.
- Ensure the dataset is optimized for exploration, modeling, and visualization.

4. Exploratory Data Analysis (EDA)

- Identify patterns and trends in the data using Python visualizations and SQL queries.
- Examine relationships between applicant characteristics and loan approval status.
- Generate insights that will guide feature selection for predictive modeling.

5. Visualization & Dashboard Development

- Plan an interactive dashboard to display insights on approval rates, risk factors, and applicant profiles.
- Include filters for exploring data by income, credit score, employment type, or region.
- Highlight actionable insights for lenders to improve decision-making.

6. Reporting & Documentation

- Compile findings, trends, and recommendations into a structured report.
- Use charts, tables, and narratives to communicate results clearly to stakeholders.

7. Project Packaging & Sharing

- Organize Python scripts, SQL files, models, dashboards, and reports into a cohesive project folder.
- Prepare a concise, portfolio-ready summary suitable for GitHub or LinkedIn